Kowndinya Boyalakuntla

Third-year Ph.D. student in Computer Science | Rutgers University

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Updated November 2026

Introduction

I am a third-year Ph.D. student in Computer Science at Rutgers University with nearly 4 years of experience in robot learning and planning. My research spans 3D scene understanding, object rearrangement, dynamic manipulation, and model-based reinforcement learning for humanoid locomotion. I have published at CoRL and IROS. My current research focuses on planning with learned world models for robust humanoid loco-manipulation.

Education

Rutgers University Sep 2024 – May 2029
Ph.D. in Computer Science; GPA: 3.85/4.0 Advisor: Prof. Abdeslam Boularias

Rutgers University Sep 2022 – Oct 2024
M.S. in Computer Science; GPA: 3.8/4.0 Advisor: Prof. Abdeslam Boularias
Thesis: Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs.

Indian Institute of Technology Tirupati Jul 2017 – Jul 2021
B.Tech. in Computer Science and Engineering; GPA: 8.42/10.0 Advisors: Prof. Sridhar Chimalakonda, Prof. Kalidas Yeturu

Research experience

Rutgers University Jan 2023 – present
Graduate Student Researcher · New Brunswick, NJ
Develop planning and model-based reinforcement learning methods for humanoid control and sim-to-real manipulation using learned world models.

RISHA Lab, IIT Tirupati — Undergraduate Researcher; Repository analysis and mobile tools. Dec 2019 – May 2022

University of Waterloo — Visiting Scholar; Large-scale analysis of 1 million GitHub repositories. Jul 2020 – Sep 2020

Publications

First-Author

1. **PL-MPC: Beyond Policy Alignment: Closing the Planning–Learning Loop for Robot Control with Learned World Models**
   Kowndinya Boyalakuntla, Yuhan Liu, Abdeslam Boularias
   Manuscript, 2026. Under review. [Paper] [Project] [Code]
   - Combined uncertainty-aware MPPI planning with value learning and policy distillation using learned world models; improved mean return over the TD-M(PC)² baseline on the challenging HumanoidBench tasks balance-hard (98 to 387) and hurdle (199 to 466), using Unitree H1 in MuJoCo.
   - Demonstrated zero-shot sim-to-real transfer of a learned world model coupled with an MPPI planner from Isaac Gym to a KUKA IIWA14 for tool manipulation on a challenging wrench–nut alignment task.
2. **Plan-Conditioned Imitation for Robust Object Retrieval under Self-Occlusion in Dense Clutter**
   Kowndinya Boyalakuntla, Ajinkya Pawar, Abdeslam Boularias, Jingjin Yu
   Manuscript, 2026. Under review. [Paper] [Project] [Code] [Data]
   - Developed a plan-conditioned imitation policy that rearranges dense clutter to create grasping clearance even when the robot's arm blocks the camera's view, with potential applications in bin picking and assembly-kit retrieval.
   - Achieved 90.7% retrieval success in Isaac Gym simulation and 90% after sim-to-real transfer to UR5e, cutting hardware task time by 65% versus the full-observation planner.
3. **KARL: Kalman-Filter Assisted Reinforcement Learner for Dynamic Object Tracking and Grasping**
   Kowndinya Boyalakuntla, Abdeslam Boularias, Jingjin Yu
   IROS, 2025 [Paper] [Code]
   - Combined curriculum reinforcement learning in Isaac Gym with Kalman-filter tracking for dynamic grasping and failure recovery through object reacquisition and grasp retries.
   - Demonstrated sim-to-real transfer to UR5e, achieving 93.3% grasp success in unobstructed scenes and 80% with obstacles, outperforming EARL in both settings.

Co-Author

1. **DAP: Diffusion-based Affordance Prediction for Multi-modality Storage**
   Haonan Chang, **Kowndinya Boyalakuntla**, Yuhan Liu, Xinyu Zhang, Liam Schramm, Abdeslam Boularias
   IROS, 2024 [Paper] [Code]
   - Collected data and evaluated diffusion-based 6D object placement for storage tasks.
   - Achieved 98% book-shelving and 94% can-stacking success in simulation, outperforming the RPDiff placement baseline; demonstrated fruit storage on a KUKA IIWA14.
2. **LGMCTS: Language-Guided Monte-Carlo Tree Search for Executable Semantic Object Rearrangement**
   Haonan Chang, Kai Gao, **Kowndinya Boyalakuntla**, Alex Lee, Baichuan Huang, Harish Udhaya Kumar, Jingjin Yu, Abdeslam Boularias
   IROS, 2024 [Paper] [Project] [Code]
   - Co-developed ELGR-Bench and implemented language parsing, object selection, and evaluation for a planner that combines language guidance with Monte Carlo tree search.
   - Achieved 79.2% executable-plan success on ELGR-Bench and demonstrated language-guided rearrangement on UR5e.
3. **Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs**
   Haonan Chang, **Kowndinya Boyalakuntla**, Shiyang Lu, Siwei Cai, Eric Pu Jing, Shreesh Keskar, Shijie Geng, Adeeb Abbas, Lifeng Zhou, Kostas Bekris, Abdeslam Boularias
   CoRL, 2023 [Paper] [Project] [Code]
   - Built the DOVE-G benchmark and evaluation pipeline for grounding language in open-vocabulary 3D scene graphs, including scene reconstruction and ConceptFusion integration.
   - Achieved 65.8% top-1 3D grounding success on DOVE-G, compared with 41.5% for OVIR-3D.

Teaching experience

Fall 2026	CS440: Introduction to Artificial Intelligence, **Course Instructor**, Rutgers University
Spring 2026	CS440: Introduction to Artificial Intelligence, Teaching Assistant, Rutgers University
Fall 2025	CS460: Introduction to Computational Robotics, Teaching Assistant, Rutgers University
Spring 2025	CS440: Introduction to Artificial Intelligence, Teaching Assistant, Rutgers University
Fall 2024	CS440: Introduction to Artificial Intelligence, Teaching Assistant, Rutgers University

Peer review

- IEEE International Conference on Robotics and Automation (ICRA) (2026)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2025, 2026)
- Conference on Robot Learning (CoRL) (2026)
- The International Journal of Robotics Research (IJRR) (2026)
- International Conference on Automated Planning and Scheduling (ICAPS) (2026)

Honors & awards

2022 Project showcase, IInvenTiv All IITs R&D Fair. Research project advised by Prof. Kalidas Yeturu, showcased at IIT Delhi.

2019 Gold Medal, Inter IIT Technical Meet 8.0. Ashoka's Tech for Change Challenge at IIT Roorkee; high-preparation event among 20 IITs.

Skills

Real hardware: UR5e, KUKA IIWA14

Simulation platforms: NVIDIA Isaac Gym, Isaac Lab, MuJoCo, PyBullet

Programming & tools: Python, PyTorch, ROS, ROS 2, NumPy, OpenCV, Slurm, Git

Leadership & mentoring

Google Developer Student Clubs, IIT Tirupati — Community Lead; 12 workshops/hackathons; 70+ students. Feb 2019 – Jul 2020

Professional service & activities

2021 Developed an Android application for IIT Tirupati's convocation.

2020 NSS, IIT Tirupati: 80 service hours; website development and maintenance team lead.